Sequential and Session-Based Recommendation

CP4285 Modern Recommendation Systems — Week 05

CP4285 Instruction Team

08 Sep 2026

Scan for the slides
soc-n.us/cp4285-t2610-w05

01 Announcements and Context

CHANGELOG

- **6 Sep 2026** — Added the Weeks 01–05 quiz-question contribution and Week 07 quiz details.
- **5 Sep 2026** — Added the Week 04 recap and Week 06 preview.



WEEK 04 RECAP

Transformer recap: query, key, value, attention

Scaled dot-product attention

input embeddings X

$Q = XW_Q$ $K = XW_K$ $V = XW_V$

$QK^\top/\sqrt{d_k}$

scaled compatibility scores

$H = \text{softmax}(QK^\top/\sqrt{d_k})V$

contextualised output embeddings

Compatibility

$$s_j = q^\top k_j$$

Queries score keys. Higher = closer match.

Weighted information

$$h = \sum_j \alpha_j v_j, \quad \alpha = \text{softmax}(s)$$

Softmax gives weights; combine values.

To think about

Week 05 foreshadowing: Transformers natively deal with ordered interactions, which we don't touch this week.

🔑 Key Point: NeuMF combines GMF + MLP

GMF

$$\mathbf{p}_u \odot \mathbf{q}_i$$

Weights coordinate-wise matches, extending the dot-product path.

NeuMF

1 · retain the GMF path

2 · add the MLP path

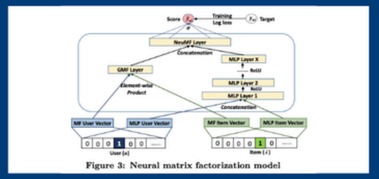


Figure 3: Neural matrix factorization model

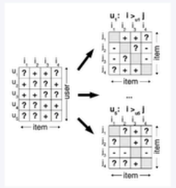
3 · concatenate both paths.

MLP

$$f(\mathbf{p}_u, \mathbf{q}_i)$$

Mixes all coordinates through learned non-linear layers.

Pointwise prediction loss vs. pairwise ranking loss



Click figure to open the source paper.
Rendle et al. (2009), Fig. 1

Pointwise: predict one interaction label

$$L_{\text{pt}} = -[r_{ui}\log\sigma(\hat{r}_{ui}) + (1-r_{ui})\log(1-\sigma(\hat{r}_{ui}))]$$

Training claim: "Did Aisha interact with City Gardens?"

Pairwise: predict one ordering

$$L_{\text{BPR}} = -\log\sigma(\hat{r}(u,i^+) - \hat{r}(u,i^-))$$

Training claim: "Should City Gardens rank above Riverside Food Court?"

The figure makes the choice visible: **pointwise calibrates a pair**; **pairwise orders a comparison**.

Rendle, S., Freudenthaler, C., Gantner, Z., & Schmidt-Thieme, L. (2009). *BPR: Bayesian personalized ranking from implicit feedback.* In *Proceedings of the Twenty-Fifth Conference on Uncertainty in Artificial Intelligence* (pp. 452–461). AUAI Press. PDF

♣ Clubs: BPR and NeuMF can both learn an attack

A pairwise win is not a robustness guarantee.

BPR-MF · loss sees

Observed positive **>** sampled unobserved item. Coordinated fake positives can create favourable training pairs for a target item.

NeuMF · capacity changes

It can fit richer interaction patterns—including an attack pattern if that pattern appears in the log. More parameters are **not** an abuse defence.

Before a deployment claim

Stress-test plausible shill or coordinated behaviour. Report ranking shifts, affected users, and target-item exposure.

If suspicious accounts or concentrated bursts are removed, does the claimed gain still hold?

⚠ Warning

Neither BPR-MF nor NeuMF is robust merely because it ranks held-out interactions well.

Quizzes - From Week 01 (10%)

- Quizzes are **jointly created by peers and instructors**
- Peers contribute candidate questions, answers, and worked solutions where needed
  - Min assigns each student a week, lecture section, difficulty, and question format.
  - Min curates, verifies, and finalises questions so that assessments remain fair, accurate, and aligned with the course learning outcomes.
- Combined worth **10%** of total marks
- **AI tools must not be used during in-class quizzes.**

> **Note**
>
> Co-creation makes quizzes part of the pilot course: students help build better learning materials for this cohort and future cohorts.



Peer Quiz Question-Bank Contribution

Covers **Weeks 01–05**. Your identifier determines one lecture section and one difficulty: **easy, medium, or hard**.

Your assigned contribution

- **Two MCQ/MRQ questions**, **two fill-in-the-blank questions**, or **one calculation question** with full working
- Include the answer for every question.

Submit on Canvas by **Mon, 21 Sep 2026, 23:59 SGT**. You have two weeks to prepare.

5 points (1% of total marks) for completion only.

Week 07 (Open Book) Quiz · Tue, 29 Sep 2026

~30 minutes at the beginning of class. A mix of instructor and peer questions, completed on paper and proctored.

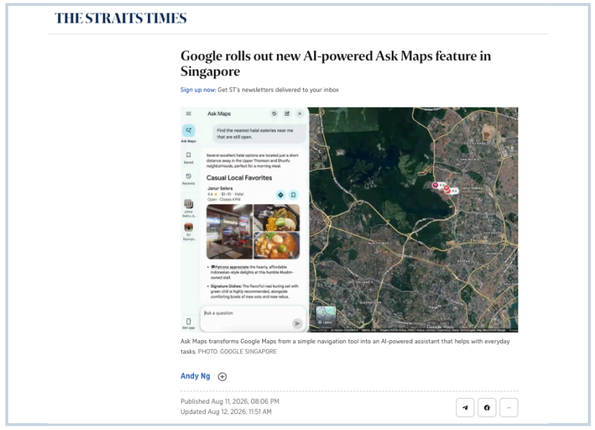


02 Lecture Overview



LAST WEEK PRE-LECTURE EXERCISE

Where Does One Session End?



THE STRAITS TIMES

Google rolls out new AI-powered Ask Maps feature in Singapore

Sign up now: Get ST's newsletters delivered to your inbox

Ask Maps

Find the nearest halal eateries near me that are still open.

Several excellent halal options are located just a short distance away in the Upper Thomson and Bhurtu neighborhoods, perfect for a morning meal.

Casual Local Favorites

Jalur Sutera

Ask a question

Ask Maps transforms Google Maps from a simple navigation tool into an AI-powered assistant that helps with everyday tasks. PHOTO: GOOGLE SINGAPORE

Andy Ng

Published Aug 11, 2026, 08:06 PM
Updated Aug 12, 2026, 11:51 AM

Optional reading: *The Straits Times*, 11 Aug 2026

Canvas discussion board · post before Mon, 7 Sep 2026 · 23:59 SGT

Post one **100–140 word** decision memo. No classmate reply is required.

> **A person asks Ask Maps for a 2pm lunch route, explores nearby cultural sites, then later asks for a halal eatery open after 10pm. Does the late-evening query begin a new session? Answer yes or no and cite two trace details.**

Next week: sequences make order, recency, and local context part of the recommendation signal.

LAST WEEK PRE-LECTURE EXERCISE · CANVAS VOICES

NO · Retain the session



“I assume all three queries … are issued sequentially in one sitting, with no idle gap.”

“The word ‘then’ implies that the query is on the same itinerary.”





“I would keep the same session while placing more emphasis on the most recent request.”

“A good recommender should avoid recommending it twice for lunch and again for supper.”



YES · Begin a new session



“Session data should only contain information … useful to the current context, whilst more general preferences … can be persisted across sessions.”

“Adding previous search context into an explicit ask is un-expected and might not be good UX.”



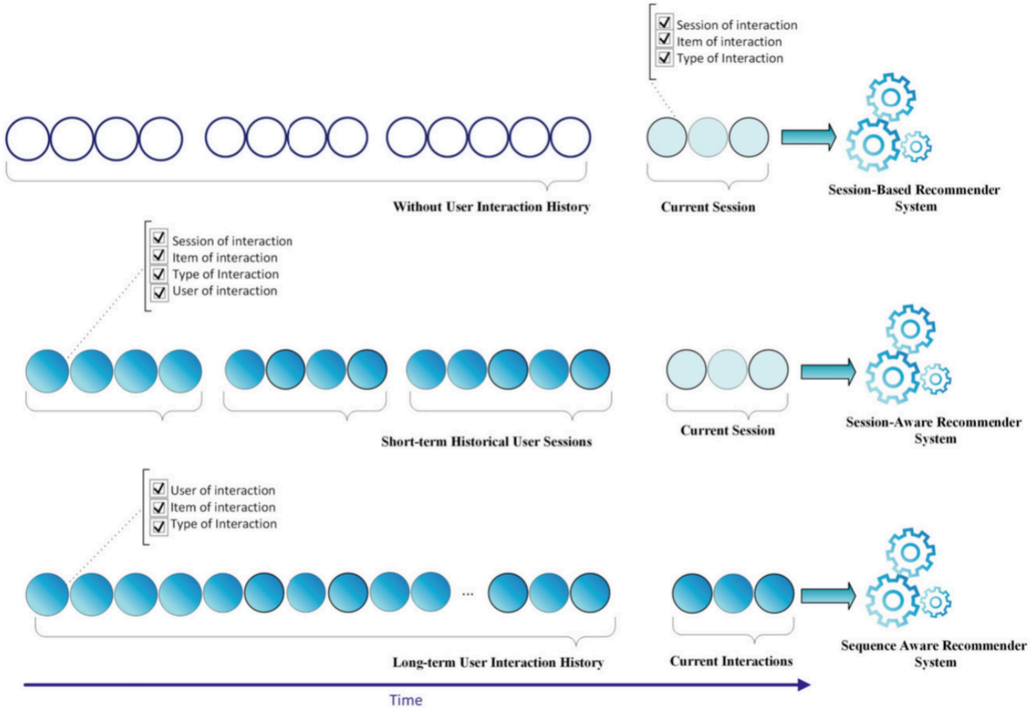


“Starting a new session does not mean discarding all earlier information.”

“That is roughly eight hours of inactivity, well beyond a typical session timeout window of thirty to sixty minutes.”



Which interaction history is available?



Session-based

Use the ordered events in the current session when no history is available.

Session-aware

Combine the current session with earlier sessions from an identified user.

Sequence-aware

Model a longer ordered history when earlier interactions may still matter.

Source: Ravanmehr, R., & Mohamadrezaei, R. (2024). *Session-based recommender systems using deep learning* (Fig. 1.3, p. 15). Springer. https://doi.org/10.1007/978-3-031-42559-2

Week 05: What changes when interactions have an order?

Week 04

Attention selected useful features for a user–item score.

Week 05

A sequence model selects useful **earlier actions** to predict the next item.

By the end of today, you should be able to:

- model temporal preferences;
- implement next-item prediction;
- compare static and dynamic representations; and
- critically assess engagement-driven objectives.



03 Session-Based Models



The immediate-intent problem

Observed now

“late-night halal food” → nearby restaurant pages → delivery times

Do not infer too much

A short trace can reveal a task without revealing a permanent identity or preference.

Session-based recommendation: predict a next item from the ordered interactions in the current, often anonymous, session.

> **Note**
>
> **“Often anonymous” includes logged-in use:** a system can keep the current session separate from the long-term profile, letting a temporary task guide the next recommendation without becoming permanent.



Bandit Time: What should this system use?

An anonymous visitor opens three restaurant pages in sequence during a late-night visit. Before recommending the next page, which should be the system's primary representation of the visitor's immediate intent?

A. A permanent "late-night diner" profile inferred from IP location and device locale

B. The ordered prefix of pages in the current session

C. An unordered bag of the three pages, ranked by citywide click-through rate

D. The highest-rated restaurant page the visitor has not opened

00:30

▶ Start Timer

Sessionisation is a modelling decision

1. Temporal context

- Timestamp and inactivity gap
- Visit duration and event pace

2. Content and task context

- Query terms and item categories
- Action types and stated constraints

3. Interaction and situational context

- Navigation path and device or browser
- Location and account continuity

Note

The boundary is an operational rule, not direct proof of intent.

To think about

A device, location, or account signal may improve next-item accuracy. What user expectation or privacy cost should limit its use?

Deciding what starts a new session?

Your group owns one session-boundary method. Apply it to the Ask Maps trace, then compare it with the shared 30-minute inactivity baseline.

Shared simplified implementation

Fixed inactivity: begin a new session after 30 minutes with no event. It is clear and cheap, but one threshold cannot fit every task.

1. Adaptive timeout

Change the gap using event pace, device, time of day, or product context.

2. Task similarity

Split when queries, items, or actions indicate a different task.

3. Boundary classifier

Learn a same-task or new-task decision from labelled traces.

4. Intent change point

Infer when the likely hidden intent state changes.



▶ Start Timer

From an event log to next-item examples

view ramen → open halal filter → inspect delivery time → **predict: late-night restaurant**

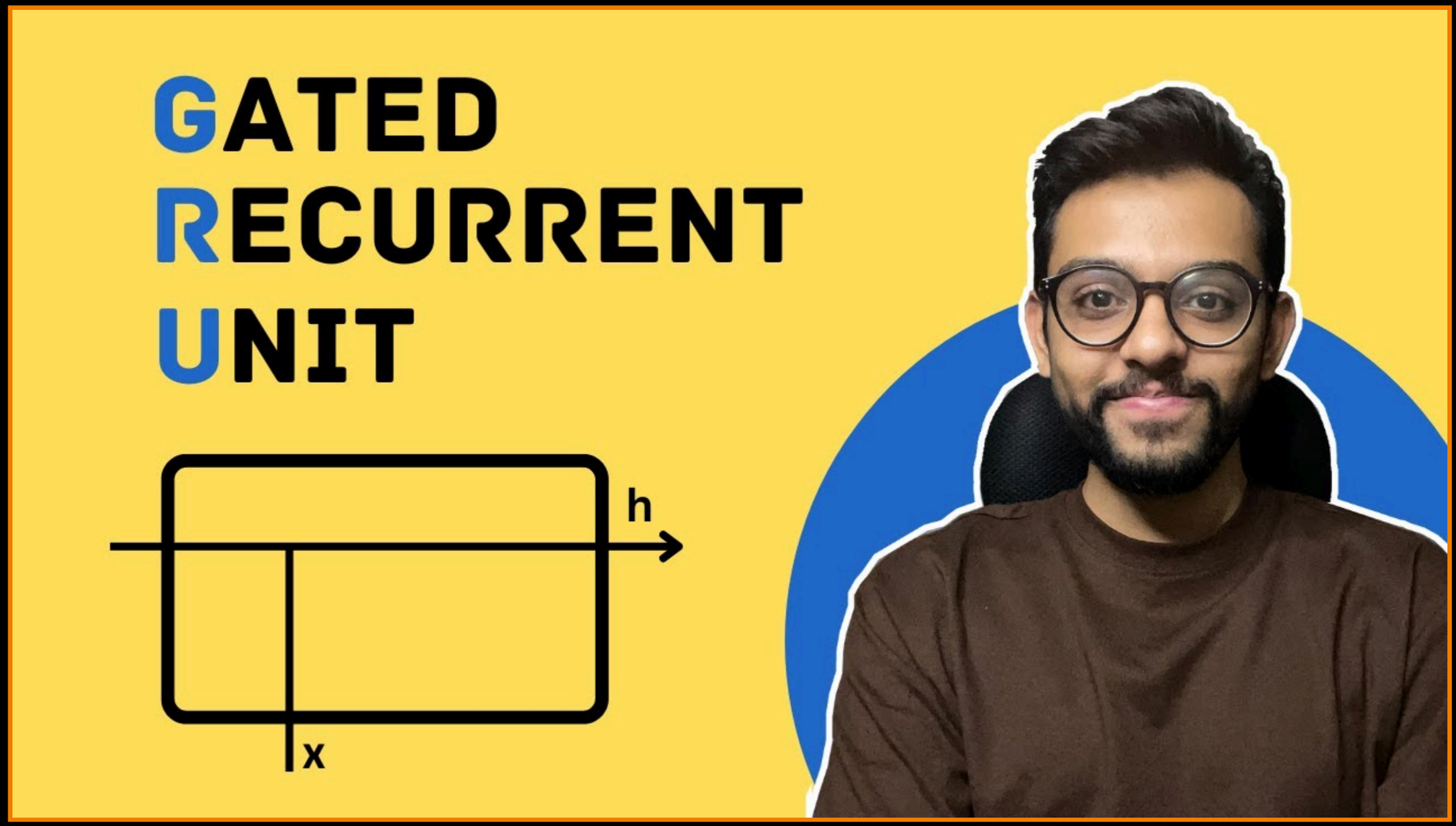
Observed prefix	Training target
[ramen]	**halal filter**
[ramen, halal filter]	**delivery time**
[ramen, halal filter, delivery time]	**late-night restaurant**

GRU4Rec

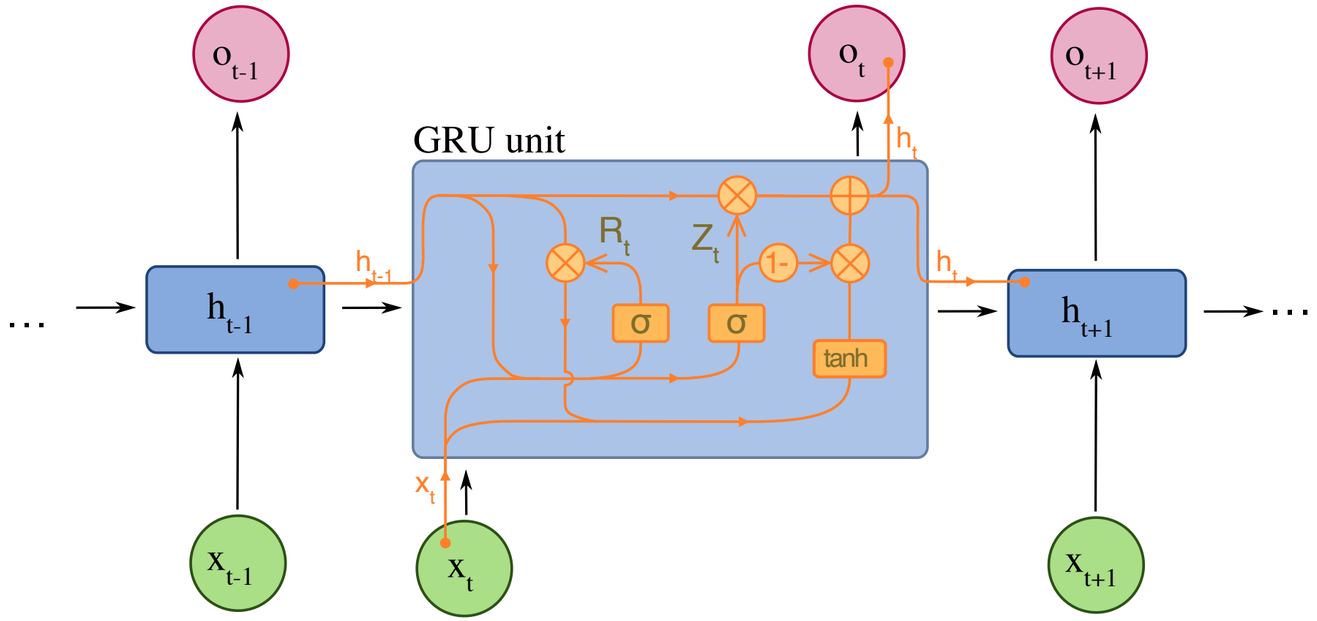
Paper: Hidasi, B., Karatzoglou, A., Baltrunas, L., & Tikk, D. (2016). *Session-based Recommendations with Recurrent Neural Networks*. ICLR.



Video: GRU intuition



GRU4Rec: carry a changing session state



Click diagram for source · fdeloche, *Gated Recurrent Unit*, CC BY-SA 4.0, via Wikimedia Commons.

Update gate z_t

Call out the final blend: how much of the previous hidden state should persist versus be replaced by the candidate state?

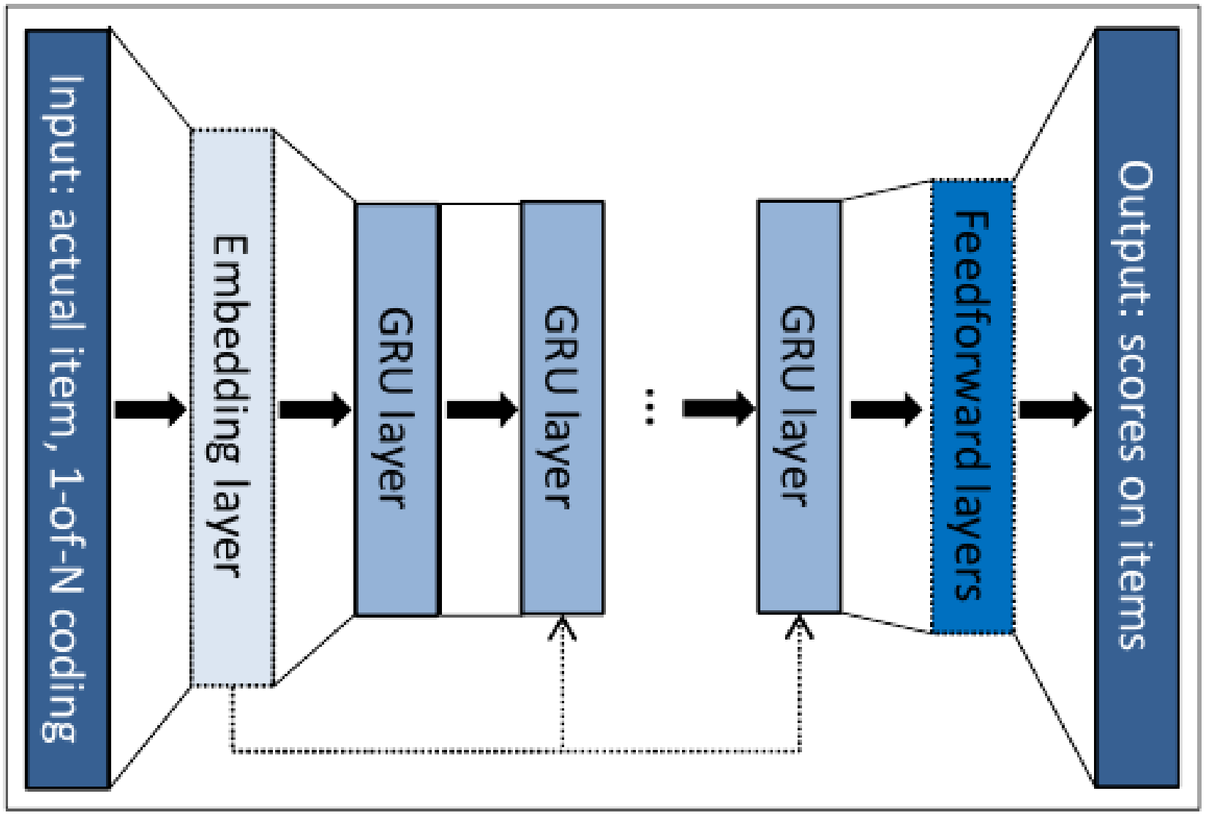
Reset gate r_t

Call out the path into the candidate state: how much earlier context should shape the next proposal?

Read the diagram

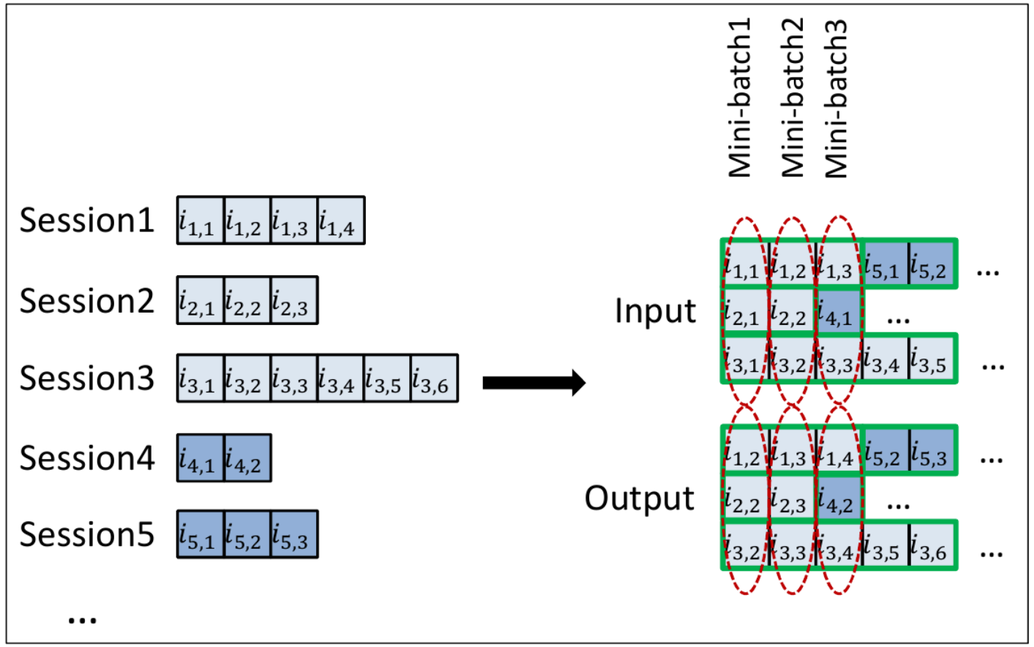
Both gates receive the current input and previous state. Their outputs shape the candidate and the final blend.

GRU4Rec architecture: one event in, scores out



Source: Hidasi, B., Karatzoglou, A., Baltrunas, L., & Tikk, D. (2016). *Session-based Recommendations with Recurrent Neural Networks* (Fig. 1, p. 4). ICLR. https://arxiv.org/abs/1511.06939

GRU4Rec implementation and evaluation lens



Batch

One next-item step per active row.

Reset on replacement

End → next session; state = zero.

Evaluate forward in time

Later sessions: HR, MRR, NDCG@K.

Source: Hidasi, B., Karatzoglou, A., Baltrunas, L., & Tikk, D. (2016). *Session-based Recommendations with Recurrent Neural Networks* (Fig. 2, p. 4). ICLR. https://arxiv.org/abs/1511.06939

GRU4Rec evaluation: BPR and TOP1

The comparison evaluated two ranking objectives: BPR and TOP1. For the observed next item ***i*** and a sampled negative ***j***, both losses want $r_i > r_j$.

BPR · push the observed item above a negative

$$L_{\text{BPR}} = -\log \sigma\left(r_i - r_j\right)$$

Only the **relative score gap** matters.

TOP1 · approximate rank, then stabilise scores

$$L_{\text{TOP1}} = \sigma\left(r_j - r_i\right) + \sigma\left((r_j)^2\right)$$

The first term penalises a negative that outranks the observed item; the second term keeps negative scores near zero.

Orange: observed next item · **Teal:** sampled negative · **Navy:** TOP1 regulariser

Note

🔑 Key Point: GRU4Rec turns a session into a next-item ranking

Use the current session to update a state, then rank what could plausibly come next.

1 · Observe order

Read the sequence of actions within one session.

2 · Update state

A GRU decides what earlier context to carry forward.

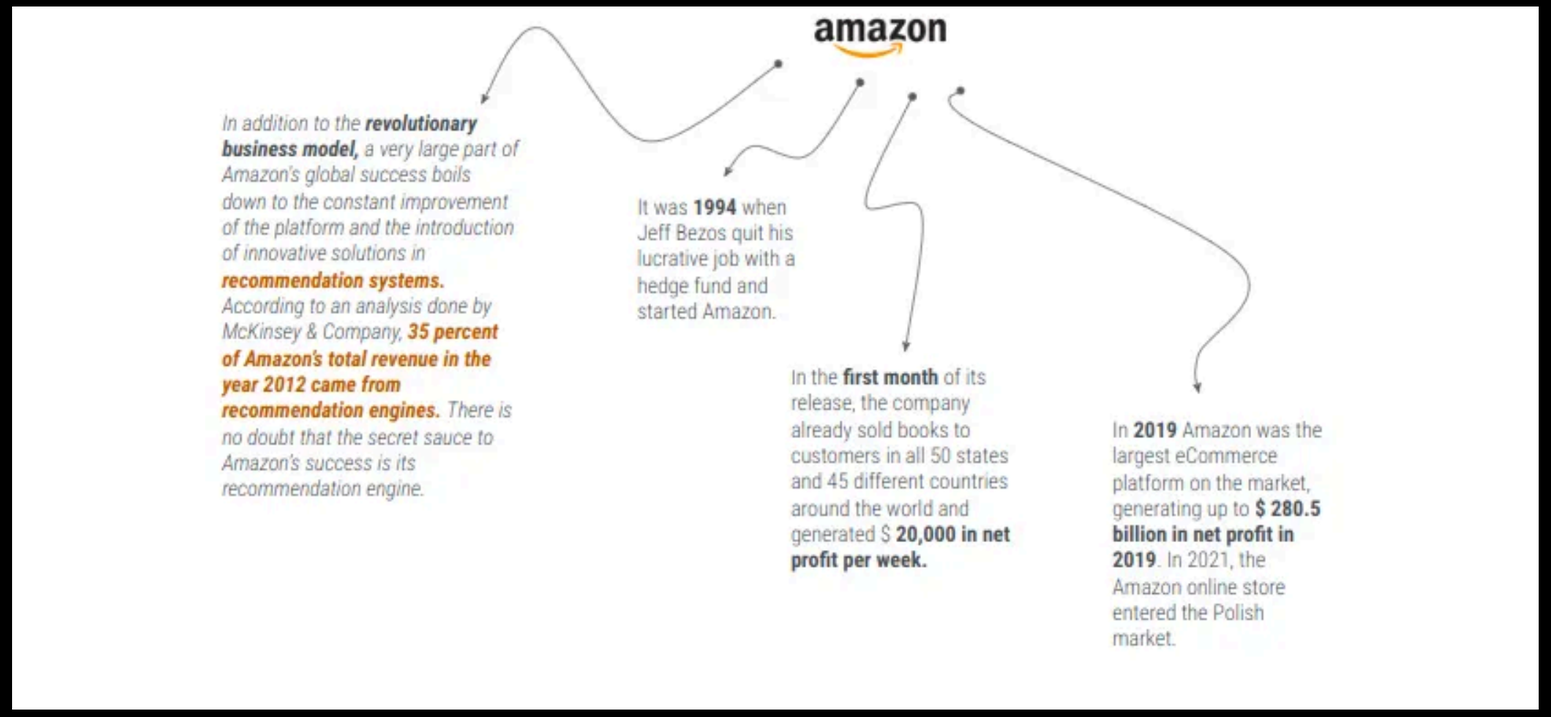
3 · Rank next items

Score candidate actions for the next step—not a permanent profile.

🤞 **Predict a next action with humility: a session state is useful, but it is not a person.**



Break — 5 minutes



WHILE YOU PAUSE

☕ BREAK

05:00

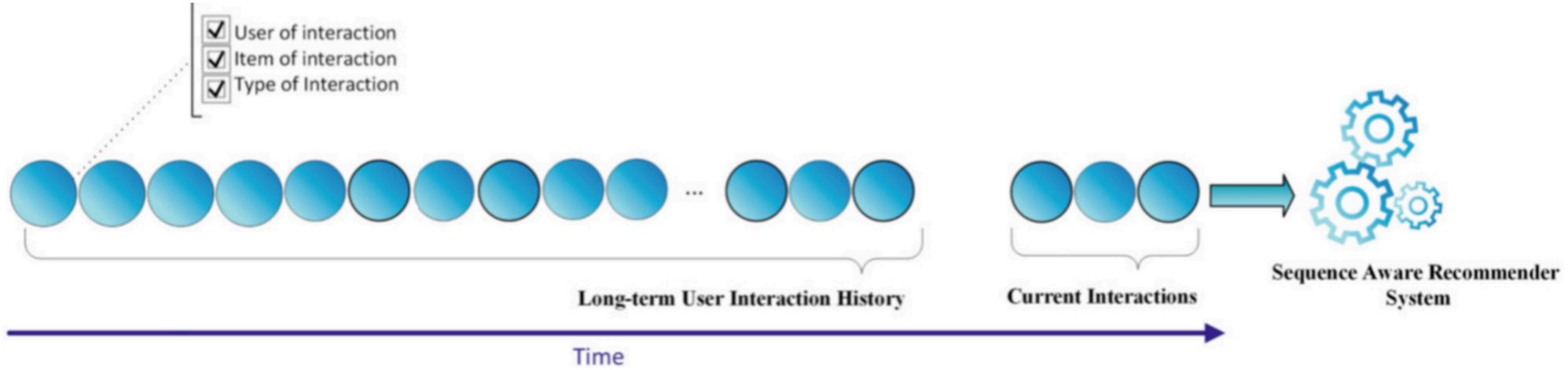
▶ Start Timer

Nicoomanesh, A. (2024, March 25). *Evolution of recommendation algorithms, Part I: Fundamentals, history overview, core and classical algorithms.* Medium. https://medium.com/@anicomanesh/evolution-of-recommendation-algorithms-part-i-fundamentals-and-classical-recommendation-bb1c0bce78a9

04 Sequential Recommendation



From the current session to a user sequence



Source: Ravanmehr, R., & Mohamadrezaei, R. (2024). *Session-based recommender systems using deep learning* (Fig. 1.3, p. 15). Springer. https://doi.org/10.1007/978-3-031-42559-2

Session state

What seems useful for the task happening now?

Sequential user state

Which earlier actions still matter after preferences evolve?

Bandit Time: Compare GRU4Rec and a new sequence model

A new causal self-attention model is proposed as an alternative to GRU4Rec for next-item ranking. Which comparison makes a fair deployment claim?

A. Use the same time cutoff, but report each model's strongest metric from its original paper.

B. Let self-attention read the full test sequence, while GRU4Rec receives only the prefix.

C. Give GRU4Rec an anonymous session but give the new model each user's complete earlier history.

D. Use one chronological cutoff; give both models only each target's preceding prefix, then use the same candidates and ranking metric.

00:45

▶ Start Timer

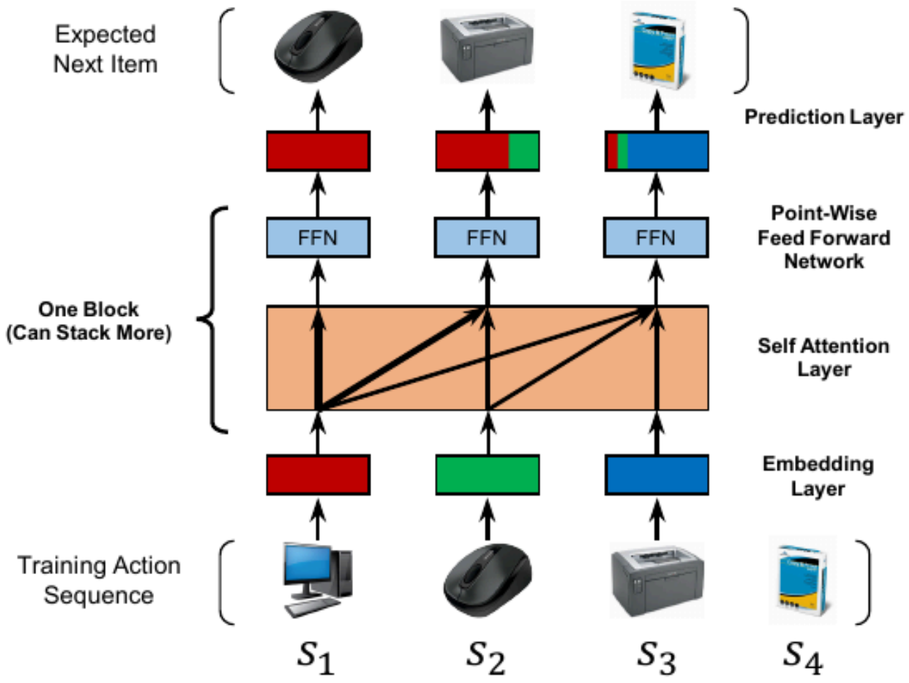
Why self-attention?

GRU4Rec — previous focus

A recurrent encoder compresses the prefix into a changing hidden state.

SASRec — current focus

Self-attention lets each prediction selectively use different earlier actions.



Source: Kang, W.-C., & McAuley, J. (2018). *Self-Attentive Sequential Recommendation* (Fig. 1, p. 1). IEEE ICDM. https://arxiv.org/abs/1808.09781

SASRec: causal self-attention for next-item prediction

item embeddings + position embeddings → masked self-attention → contextual sequence representation → score next items

Causal mask

At position *t*, the model cannot inspect future actions.

Position matters

The same items in a different order can imply a different next item.

SASRec walkthrough: one next-item recommendation at a time

Causal self-attention over the available session prefix

museum ticket	vegetarian café	late-night halal food
earlier	earlier	current query

Step 1 · attention scores

0.08 0.22 **0.70**

Step 2 · causal softmax weights

8% 22% **70%**

Step 3 · contextual state

The model gives the late-night halal action the largest weight, while retaining the earlier context.

Ranked next-item recommendations

Session prefix
museum ticket
→ vegetarian café
→ late-night halal food

1. delivery-time check
highest-ranked next item

2. halal restaurant details
re-rank after the new observed action

3. order tracking
re-rank again with the longer prefix

Only past actions are visible.

Preserve chronology in the learning pipeline

earlier history → train sequence model → later held-out interaction → rank candidates

A random split leaks the future

It lets the model learn from actions that had not happened at recommendation time.

Causal masking applies the same rule within a sequence

Before softmax, set scores for future positions to −∞. Each prediction can use its prefix, never a later action.



Source: Jay Alammar, *The Illustrated GPT-2* (causal attention-mask diagram). jalammar.github.io/illustrated-gpt2

Transformer recommenders: related tasks, different masks

Model framing	Training visibility	Natural use
SASRec	Causal: past only	**Next-item prediction**
BERT4Rec-style	Masked positions with bidirectional context	**Masked-item training; adapted to next-item ranking at inference**

Worked trace: what should the model attend to?

Trace

vegetarian café → museum ticket → “late-night halal food” → delivery-time check → ?

Name one action likely to be highly relevant, one action that may be weakly relevant, and one uncertainty that a log alone cannot resolve.



👥 Long-term profile or sharp current intent?

For the worked trace, decide what should influence the next recommendation, what must not overwrite the profile, and one non-click outcome that would show helpfulness.

Write a two-sentence answer. Be ready to name one temporal signal, one uncertainty, and one risk of optimizing only engagement.

03:00

▶ Start Timer



Choose the smallest sufficient temporal model

Evidence and need	Reasonable starting point
One immediately preceding action is enough	Markov / co-occurrence baseline
Short, changing anonymous intent	GRU4Rec or session-aware method
Long ordered history with selective dependencies	SASRec-style sequential Transformer



05 Summary



🔑 Time changes what a preference signal means.

Session

Local intent can be useful without defining a person.

Sequence

Order and recency can change which history matters.

Objective

A next click is not the whole definition of help.

NEXT WEEK PRE-LECTURE EXERCISE

When should the current session outweigh history?

> **A user’s long-term history is vegetarian cafés. In their current session, they search “late-night halal food”, open nearby restaurant pages, and check delivery times. What should the app consider now—and what should it not assume?**

Canvas: 120–140 words by Mon, 14 Sep 2026, 23:59 SGT. Name two session-relevant item types, one risk of overwriting durable preference, and one non-click helpfulness signal.

Next week: retrieve plausible options before ranking and re-ranking them.



NEXT WEEK PREVIEW — WEEK 06

Retrieval and Ranking Architectures

A production recommender cannot compare every catalogue item in one step. It first **retrieves** plausible candidates, then **ranks** and re-ranks the short list for the user and situation.

In Week 06:

- trace the candidate-generation → ranking → re-ranking pipeline;
- compare retrieval and ranking objectives; and
- examine two-tower architectures and visibility trade-offs.



Photo: Barnabas Davoti / Unsplash